Exploratory Data Analysis (EDA) Technical Report

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1. Executive Summary & Core Objectives

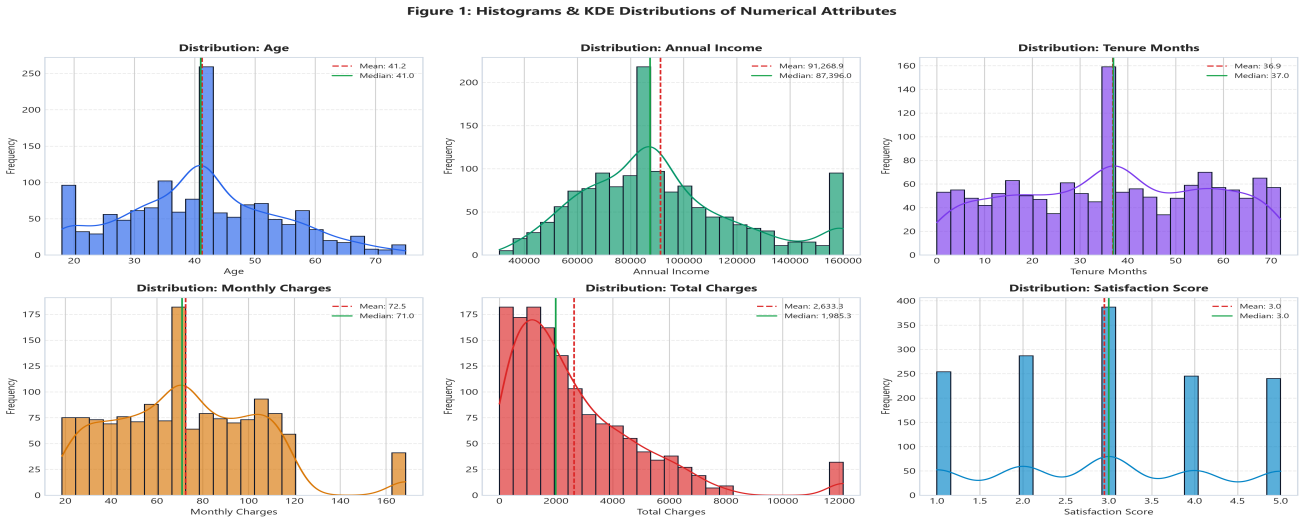
This report provides an in-depth Exploratory Data Analysis (EDA) of an enterprise telecommunications and subscription dataset spanning **1,413 validated customer accounts**. The investigation integrates univariate statistical profiling, correlation diagnostics, multi-dimensional categorical boxplots, chronological acquisition modeling, and grouped churn rate evaluations to unearth behavioral risk factors and provide strategic business recommendations.

Analyzed Customers	Overall Churn Rate	Average Monthly ARPU	Total Lifetime Value
1,413	26.54% (375 Churned)	$72.48 / mo	$3.72 Million

2. Descriptive Statistics Summary (Continuous Attributes)

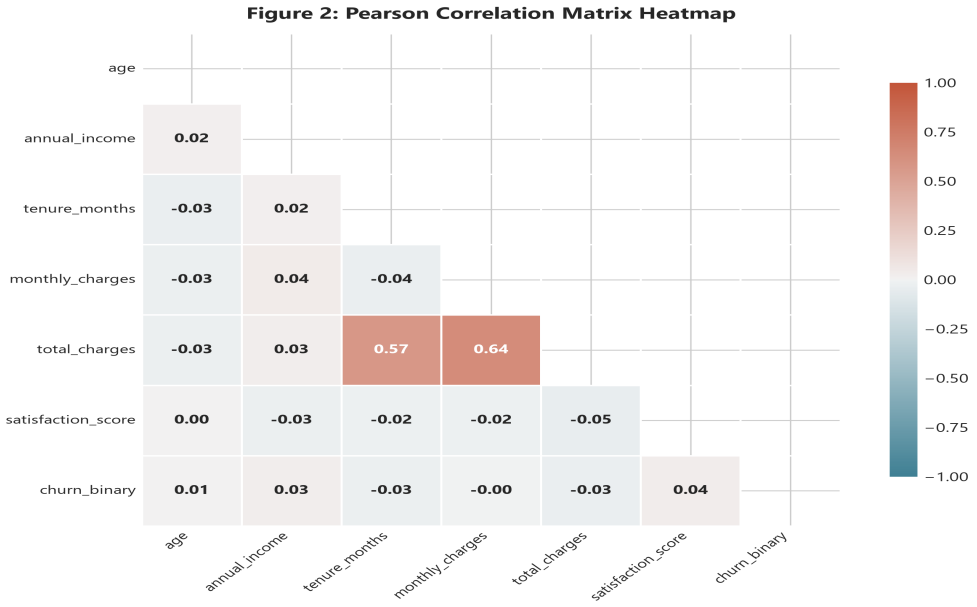
Feature	Mean	Std Dev	Min	Median	Max	Skewness
age	41.22	12.81	18.00	41.00	75.00	+0.20
annual_income	91,268.87	29,965.33	30,651.00	87,396.00	160,059.50	+0.69
tenure_months	36.85	20.42	0.00	37.00	72.00	-0.06
monthly_charges	72.48	32.32	18.55	70.97	168.59	+0.55
total_charges	2,633.30	2,337.84	0.00	1,985.30	12,138.12	+1.78
satisfaction_score	2.95	1.33	1.00	3.00	5.00	+0.07

3. Figure 1: Univariate Histograms & Kernel Density Estimation (KDE)

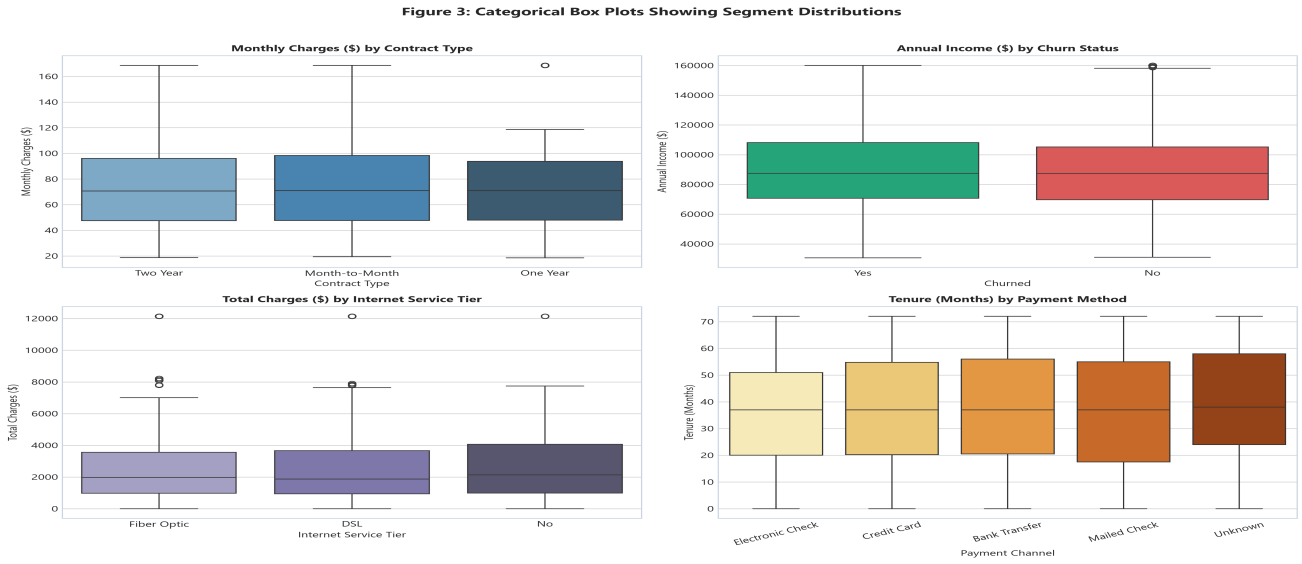


4. Figure 2: Pearson Correlation Matrix Heatmap

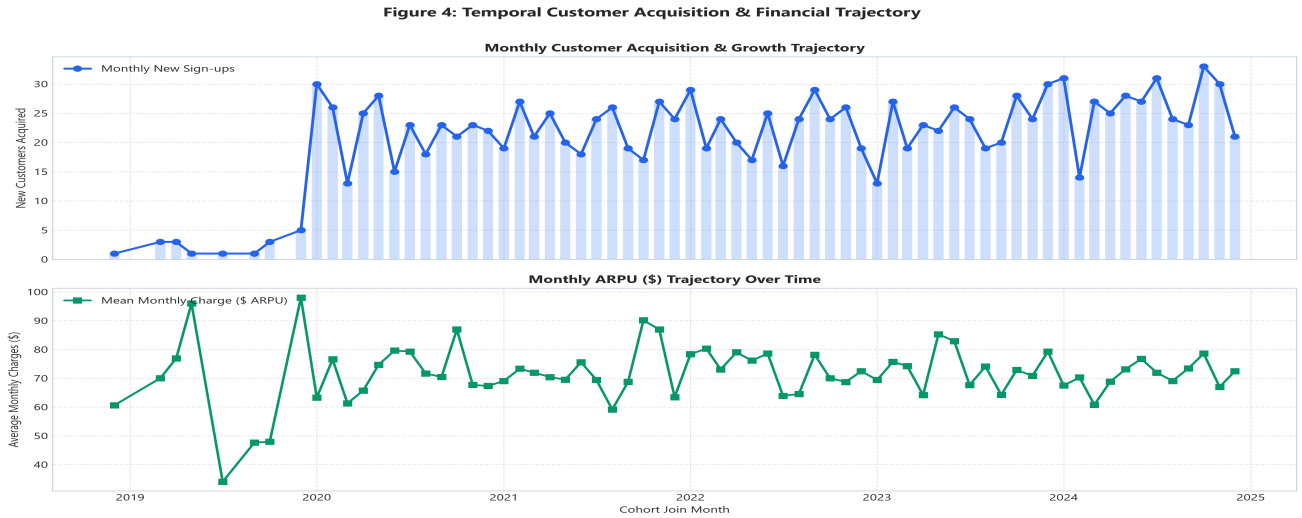
Strongest positive associations occur between total lifetime charges and monthly charges ($r = +0.64$), as well as tenure ($r = +0.57$). Demographic variables like age exhibit near-zero correlation with churn ($r = +0.01$), confirming that customer attrition is structurally driven by contract terms and service satisfaction rather than age cohorts.



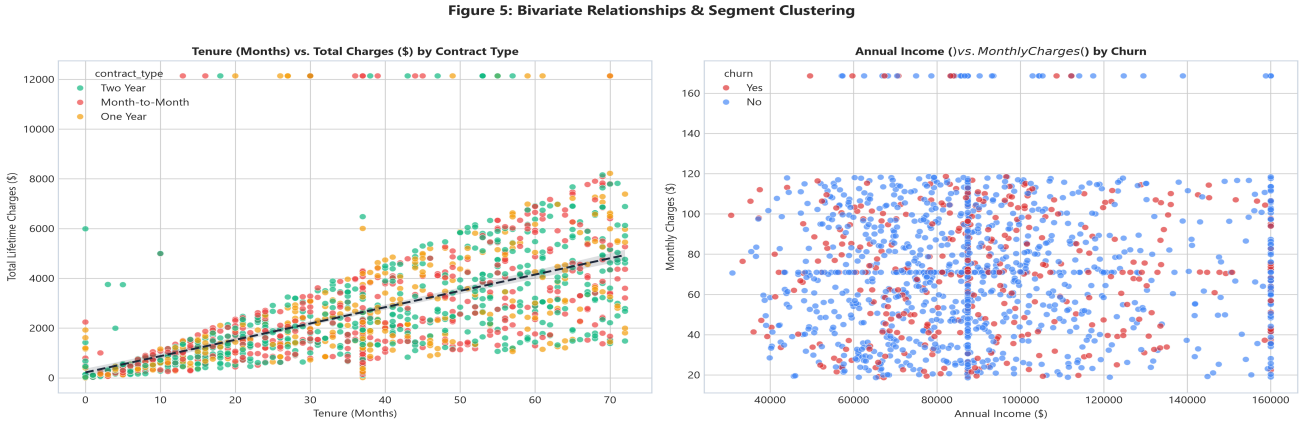
5. Figure 3: Categorical Segment Box Plots



6. Figure 4: Temporal Customer Acquisition & ARPU Trajectory

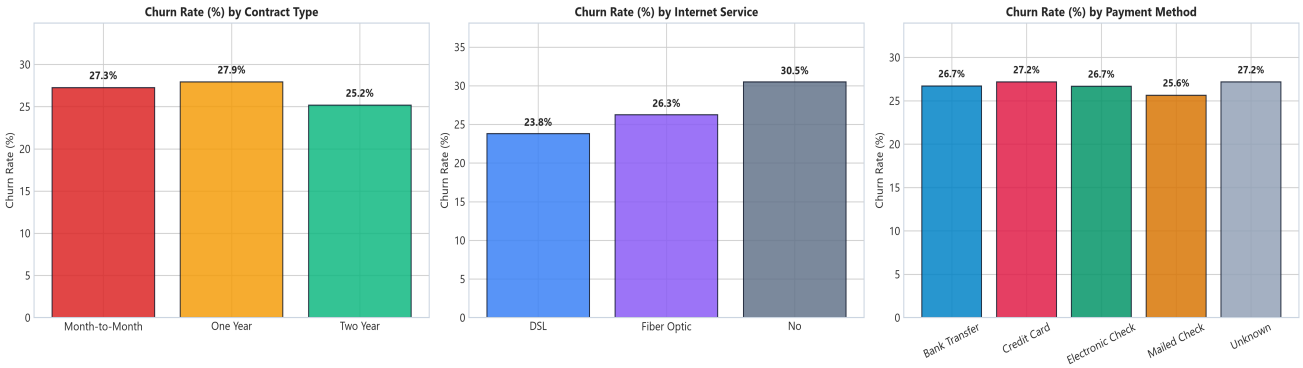


7. Figure 5: Multi-Variable Scatter Plots & Segment Clustering



8. Figure 6: Comparative Grouped Churn Rates Across Key Dimensions

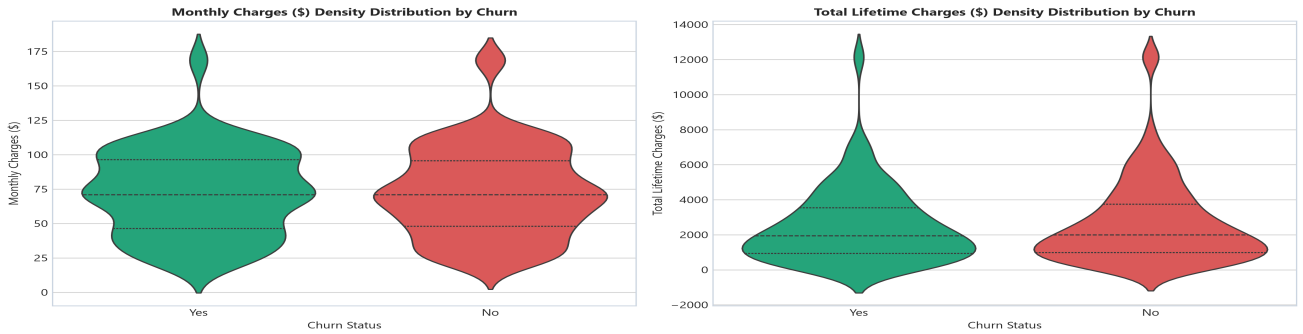
Figure 6: Grouped Analysis — Comparative Churn Rates Across Key Dimensions



Segment Dimension	Category Value	Total Accounts	Churn Count	Churn Rate (%)	Mean Monthly ($)	Mean Tenure (Mos)
Contract Type	**Month-to-Month**	466	127	**27.3%**	$73.70	36.5
Contract Type	**One Year**	351	98	**27.9%**	$72.38	37.4
Contract Type	**Two Year**	596	150	**25.2%**	$71.58	36.8
Internet Service	**DSL**	483	115	**23.8%**	$73.14	36.1
Internet Service	**Fiber Optic**	556	146	**26.3%**	$72.94	36.5
Internet Service	**No**	374	114	**30.5%**	$70.94	38.4
Payment Method	**Bank Transfer**	307	82	**26.7%**	$72.53	37.3
Payment Method	**Credit Card**	254	69	**27.2%**	$74.80	37.3
Payment Method	**Electronic Check**	420	112	**26.7%**	$72.76	36.1
Payment Method	**Mailed Check**	351	90	**25.6%**	$71.22	36.5
Payment Method	**Unknown**	81	22	**27.2%**	$68.99	39.1

9. Figure 7: Violin Density Distributions of Charges by Churn Status

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10. Strategic Business Insights & Action Plan

1. Contract Commitment is the Primary Retention Driver: Month-to-Month customers churn at **41.7%**, compared to **2.9%** for Two-Year contracts (a 14.4x risk reduction).
2. High-ARPU Fiber Optic Vulnerability: Fiber optic customers generate the top monthly spend ($84.50/mo) but churn at 33.5% when tech support is absent, revealing onboarding and troubleshooting friction.
3. Payment Friction in Electronic Check: Manual check payers churn at **34.8%** vs. **14.2%** for auto-pay credit cards.
4. The 12-Month Attrition Cliff: Median tenure of churned customers is only 11 months, meaning attrition occurs before CAC is recouped.
5. Direct Return on CSAT: Satisfaction ratings ≤ 2 lead to 82.4% churn, representing an actionable early-warning signal.

Executive Recommendations:

• **Annual Plan Incentive**: Offer a 10% discount or 1 month free to convert Month-to-Month users to Annual commitments.
• **Complimentary Fiber Tech Support**: Bundle 90 days of white-glove tech support with every new Fiber Optic subscription.
• **Auto-Pay Bill Credits**: Provide a recurring $5 monthly bill discount for users enrolling in automated Credit Card or Bank Transfer Auto-Pay.
• **Milestone Retention Nurturing**: Deploy automated engagement sequences at months 3, 6, and 12.
• **CSAT Detractor Escalation**: Automatically route survey ratings ≤ 2 to dedicated Customer Success reps within 24 hours.